

Riyasat Ohib

CONTACT
INFORMATION

Google DeepMind
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 GitHub
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 Google Scholar

RESEARCH
INTERESTS

My research examines the computational and representational organization of deep neural networks. I study how internal structure develops through learning and shapes reasoning, adaptation, and model behavior, using efficient methods that extend to large models. A central interest is alignment, especially how stable patterns of behavior and model personas form and persist, and how they can be controlled. I am also deeply interested in the science of human and machine learning.

EDUCATION

PhD, Georgia Institute of Technology Aug 2021 – May 2026
Electrical and Computer Engineering Atlanta, GA
Concentration in Artificial Intelligence · GPA: 4.0/4.0
Dissertation: The Hidden Structure of Deep Models: Discovering and Exploiting Sparse Subnetworks
Advisors: [Vince D. Calhoun](#) and [Sergey Plis](#)

MS, Georgia Institute of Technology Aug 2019 – Aug 2021
Electrical and Computer Engineering Atlanta, GA
Concentration in Artificial Intelligence · GPA: 4.0/4.0
Thesis: Explicit Group Sparse Projection for Machine Learning

EXPERIENCE

Google DeepMind · Research Scientist Jul 2026 – Present
Permanent, full-time New York City, NY

- Research on representations, control, alignment and interpretability of large-scale frontier models.

Google DeepMind · Research Intern Sep 2025 – May 2026
Model Alignment, Control, Responsibility and Safety Kirkland, WA

- Studied how concepts are represented in the residual stream of diffusion transformers, and how that structure can be used to steer generation (CVPR, HOW 2026).

Cohere · Intern of the Technical Staff Sep 2024 – Dec 2024
LLM Efficiency Research Atlanta, GA

- Developed inference-time activation sparsity methods for large language models, targeting deployment settings where latency and serving cost dominate.

Dolby Laboratories · PhD Research Intern May 2024 – Aug 2024
Experience Delivery Lab, Advanced Technologies Group Atlanta, GA

- Designed a parameter-efficient fine-tuning method for LLMs based on probabilistic layer selection, concentrating updates on the layers most relevant to the downstream task.

Meta AI (FAIR) · Research Scientist Intern May 2022 – Aug 2022
Sparsity and Efficiency Menlo Park, CA

- Designed and implemented *WeiGit*, a git-like library for versioning and compressing neural network weights during training; released as part of facebookresearch/fairscale.
- Studied extreme sparsity in deep networks using signal-processing transforms (FFT, DCT) applied during training.

EXPERIENCE
(continued)**TReNDS Center** · Graduate Research Assistant

Aug 2019 – Aug 2026

Georgia Institute of Technology

Atlanta, GA

- Developed Grouped Sparse Projection, a projection-based algorithm for training sparse deep models that retains near-baseline accuracy above 90% sparsity (TMLR 2022; ICLR HAET workshop 2021).
- Developed communication-efficient federated learning methods for the non-IID regime, with applications to distributed neuroimaging (TMLR 2026; *Frontiers in Neuroinformatics* 2024).
- Developed single-shot pruning and neural-pathway discovery methods for offline and multi-task reinforcement learning, in collaboration with [Mila](#), Montreal (NeurIPS 2024).
- Introduced sparse adapters as parameter-efficient building blocks for modular architectures, outperforming LoRA and full fine-tuning when merged across up to 20 NLP tasks, in collaboration with Mila (COLM 2025).

PUBLICATIONS

* Equal contribution

- [1] **R. Ohib**, M. Hahn, M. Malek. *Concept Spaces in the Residual Stream of Diffusion Transformers*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), HOW, 2026. [\[paper\]](#)
- [2] **R. Ohib**, B. Thapaliya, G. K. Dziugaite, J. Liu, V. D. Calhoun, S. Plis. *SSFL: Discovering Sparse Unified Subnetworks at Initialization for Efficient Federated Learning*. Transactions on Machine Learning Research (TMLR), 2026. [\[paper\]](#)
- [3] J. Gammell, B. Thapaliya, Y. Jung, **R. Ohib**, B. Fehri, D. Chakrabarti. *Learning to Query History: Nonstationary Classification via Learned Retrieval*. ICLR Workshop on Time Series in the Age of Large Models, 2026. [\[paper\]](#)
- [4] S. Y. Arnob, Z. Su, M. Kim, O. Ostapenko, **R. Ohib**, E. Saleh, D. Precup, L. Caccia, A. Sordoni. *Exploring Sparse Adapters for Scalable Merging of Parameter Efficient Experts*. Conference on Language Modeling (COLM), 2025. [\[link\]](#)
- [5] S. Y. Arnob, **R. Ohib**, S. Plis, A. Zhang, A. Sordoni, D. Precup. *Efficient Reinforcement Learning by Discovering Neural Pathways*. Advances in Neural Information Processing Systems (NeurIPS), 2024. [\[link\]](#)
- [6] B. Thapaliya*, **R. Ohib***, E. Geenjaara, J. Liu, V. D. Calhoun, S. Plis. *Efficient Federated Learning on Distributed Neuroimaging Data*. Frontiers in Neuroinformatics, 2024. [\[link\]](#)
- [7] E. Geenjaara, D. Kim, **R. Ohib**, M. Duda, A. Kashyap, S. Plis, V. D. Calhoun. *Uncovering the Latent Dynamics of Whole-Brain fMRI Tasks with a Sequential Variational Autoencoder*. NeurIPS Deep Generative Models for Health Workshop, 2023. [\[link\]](#)
- [8] B. Thapaliya, **R. Ohib**, E. Geenjaara, J. Liu, V. D. Calhoun, S. Plis. *Decentralized Sparse Federated Learning for Efficient Training on Distributed Neuroimaging Data*. NeurIPS Medical Imaging Workshop, 2023. [\[link\]](#)
- [9] **R. Ohib**, B. Thapaliya, P. Gaggenapalli, J. Liu, V. D. Calhoun, S. Plis. *SalientGrads: Sparse Models for Communication Efficient and Data Aware Distributed Federated Training*. ICLR Sparsity in Neural Networks Workshop, 2023. [\[arXiv\]](#)
- [10] **R. Ohib**, N. Gillis, N. Dalmasso, S. Shah, V. Potluru, S. Plis. *Explicit Group Sparse Projection with Applications to Deep Learning and NMF*. Transactions on Machine Learning Research (TMLR), 2022. [\[link\]](#)
- [11] S. Y. Arnob, **R. Ohib**, S. Plis, D. Precup. *Single-Shot Pruning for Offline Reinforcement Learning*. NeurIPS Offline Reinforcement Learning Workshop, 2021. [\[arXiv\]](#)
- [12] **R. Ohib**, N. Gillis, S. Shah, V. Potluru, S. Plis. *Grouped Sparse Projection for Deep Learning*. ICLR Hardware Aware Efficient Training Workshop, 2021. [\[link\]](#)
- [13] **R. Ohib**, S. Y. Arnob, M. Muhaisin, R. Arefin, T. Reza, M. R. Amin. *ENF Based Machine Learning Classification for Origin of Media Signals: Novel Features from Fourier Transform Profile*. International Conference on Electrical, Electronics and Computer Science (ICEECS), 2018.
- [14] S. Y. Arnob, **R. Ohib**, M. Muhaisin, T. B. Hassan. *Power File Extraction Process from Bangladesh Grid and Exploring ENF Based Classification Accuracy Using Machine Learning*. IEEE Region 10 Humanitarian Technology Conference (R10-HTC), 2017. [\[link\]](#)

HONORS AND AWARDS	2019	CIFAR and Amii travel grant, Deep Learning and Reinforcement Learning Summer School (DLRLSS), Edmonton, Canada
	2016	Honorable Mention and 11th place worldwide, IEEE Signal Processing Cup (team and programming lead)
	2013 – 2017	Undergraduate Honors List, IUT
	2013 – 2017	OIC Scholarship — full tuition waiver and monthly stipend for the duration of undergraduate study
TALKS AND PRESENTATIONS	Jul 2026	<i>The Hidden Structure of Deep Models: Discovering and Exploiting Sparse Subnetworks</i> , TReNDS Center, Atlanta, GA.
	Apr 2026	<i>What Models Learn and What They Hide: Sparse Structures and Representation Control</i> , Google DeepMind, Kirkland, WA.
	Apr 2026	<i>Navigating a PhD and Doing Scientific Research</i> , TReNDS Center, Atlanta, GA.
OPEN SOURCE	WeiGit	2022
	Weight versioning for neural networks · Meta AI (FAIR) · facebookresearch/fairscale	
	• Version control and compression for model weights during training, exposed through a git-like command line and Python API; merged into the fairscale library.	
ACADEMIC SERVICE	2026	Program Committee , International Semantic Web Conference (ISWC)
	2023 – Present	Reviewer Transactions on Machine Learning Research (TMLR) Conference on Neural Information Processing Systems (NeurIPS) International Conference on Learning Representations (ICLR) IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
GRADUATE COURSEWORK	• Statistical Machine Learning • Convex Optimization • Linear Algebra • Advanced Digital Signal Processing • Advanced Programming Techniques • Fourier Analysis • Information Processing in Neural Systems • Real Analysis	
PROGRAMMING	Languages	Python, C++, Matlab, Bash
	Frameworks	PyTorch, JAX, NumPy, Pandas
	Infrastructure	Linux, Slurm, distributed and multi-GPU training, cluster computing